

Week 4

Incremental cash flow

This week we will:

4.1: Incremental Cash Flows in a capital budgeting SLO1, 2 & 4

4.2: Special Cash Flow items SLO1, 2 & 4

4.3: Taxation and depreciation SLO1, 2 & 4

4.4: Capital budgeting process SLO1, 2 & 4

4.5: A Capital budgeting example SLO1, 2 & 4

Learning objectives

At the end of this module you will:

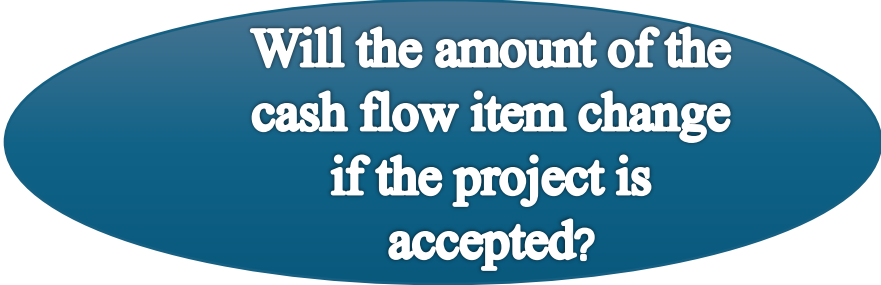
SLO1: describe the principles and objectives of financial decision-making

SLO2: apply time-value of money techniques to the valuation of financial instruments and financial securities

SLO4: evaluate a corporate investment proposal and provide a recommendation to accept or reject it

What is an incremental cash flow?

- Incremental cash flows are **changes in cash flows** that occur from accepting the project
- To determine if a cash flow will be incremental ask yourself:



**Will the amount of the
cash flow item change
if the project is
accepted?**

A cash flow that occurs regardless of whether an investment proceeds or does not proceed is not incremental and is excluded

Practice 1

- Your firm's annual sales are currently \$3million a year. As CFO you are examining a new project that will increase your sales to \$5 million. What is the incremental cash flow?
 - a) \$5 million
 - b) \$3 million
 - c) \$8 million
 - d) \$2 million

Answer: D

Practice 2

- A company with annual sales of \$100 million plans to introduce a new product. However, it will compete with their existing products
 - New product sales are expected to be \$25m pa
 - But existing product sales will fall by \$7m pa
-
- What is the incremental sales figure?
- +\$18M

Practice 3

- A retailer currently with 15 stores wants to open a new store in Canberra. The CEO wants to include the new store's share of head office costs ($1/16^{\text{th}}$) in the evaluation of the project. This amounts to \$22,500 per year. The opening of the Canberra Store will not increase head office costs.
- Should the \$22,500 be included in the evaluation?
No, it is not incremental to the company.

Capital Budgeting Consideration

1. Identify cash flows

2. Include or Ignore?

- Sunk costs?
- Opportunity costs?
- Taxes?

3. Calculate incremental cash flows

4. Where is the cash flow?

- Start / Life / End of project?

5. Calculate NET cash flow

6. Determine required rate

Calculate NPV

Special cash flow items—Sunk costs

- A cost is **incurred regardless of the investment decision**
- Is an expense **paid previously** by the firm and cannot be recovered (sunk).
 - Eg: A company paid \$50,000 on market research into Diet Tim Tams. The \$50,000 was spent regardless of the final decision.
- Only relevant cash flows are incremental cash flows
- **IGNORE sunk costs in project analysis!!**

Special Cash Flow Items—Opportunity cost

- **Cash flows forgone (given up) due to using an asset for the project**
 - A firm plans to build a factory on land it owns.
 - The firm loses the opportunity to use the land for an alternative purpose and forgoes the proceeds (e.g. selling the land)
- We use the most valuable alternative to measure opportunity cost (**market value**)
- **Opportunity Cost is an incremental cash flow!**

Special Cash Flow Items—Financing cost

- Finance Costs refer to the costs that you need to pay to firm's debtholders and shareholders in order to fund/finance the project.
- For example:
 - Financing costs paid to debtholders: Interest expense
 - Financing costs paid to shareholders: Dividend
- **Interest payments and other financing cash flows should NOT be included in our capital budgeting analysis**

Special Cash Flow Items—Working Capital item

- Managing short term assets and liabilities ...
- **Current Assets** –liquid securities, inventory, accounts receivable
- **Current Liabilities** – accounts payable
- **Working capital items need to be included at the start of our project and fully reversed in the last year of the projec**

Special Cash Flow Items—Working Capital item

How are the working capital items affecting the incremental cash flows?

An increase in Current Asset	→	Cash Outflow (-)
A decrease in Current Asset	→	Cash Inflow (+)
An increase in Current Liability	→	Cash Inflow (+)
A decrease in Current Liability	→	Cash Outflow (-)

Practice 1

- Identify whether or not the following cash flows need to be included in our capital budgeting analysis? If so, what is the incremental cash flow amount?

Example	Included in our capital budgeting analysis?	If included, what is the incremental cash flow amount? (Include the cash flow)
Due to the increased productivity, the revenue increases from \$10,000 to \$40,000		
Due to the increased productivity, the cost is decreased by \$40,000		
The fixed cost of the firm remains at \$30,000, regardless of the investment decision		
The firm pays \$30,000 of interest expense per year		
Instead of selling a machine to make a gain of \$30,000, the firm chooses to keep it for the new project		
Market research costing \$20,000 was conducted prior to the start of the project		
There is a need to increase the accounts payable by \$10,000 to meet the cash flow need		

Practice 1

- Identify whether or not the following cash flows need to be included in our capital budgeting analysis? If so, what is the incremental cash flow amount?

Example	Included in our capital budgeting analysis?	If included, what is the incremental cash flow amount? (Include the cash flow)
Due to the increased productivity, the revenue increases from \$10,000 to \$40,000	Yes-Incremental cash flow (from \$10,000 to \$40,000)	+30,000
Due to the increased productivity, the cost is decreased by \$40,000	Yes-Incremental cash flow (cost is saved by \$40,000)	+40,000
The fixed cost of the firm remains at \$30,000, regardless of the investment decision	No-No incremental cash flow (whether the project is invested or not the fixed cost still needs to be paid)	
The firm pays \$30,000 of interest expense per year	No-Financing cost	
Instead of selling a machine to make a gain of \$30,000, the firm chooses to keep it for the new project	Yes-Opportunity Cost. The loss of making a gain of \$30,000 right now by selling the machine due to the new project	-\$30,000
Market research costing \$20,000 was conducted prior to the start of the project	No- Sunk cost	
\$10,000 worth of inventory was sold at the start of the project due to no demand	Yes-Working Capital	+10.000
There is a need to increase the accounts payable by \$10,000 to meet the cash flow need	Yes-Working Capital	+10.000

Incremental cash flow

- ☐ A retail firm with 15 branches is to open a new branch in Jindabyne
- ☐ As part of the project evaluation the CEO wants to include one sixteenth of the head office costs in the Jindabyne expenses
- ☐ This amounts to \$22,500 per year.
- ☐ The opening of the Jindabyne branch will not increase head office costs.
- ☐ Should the \$22,500 be included in the evaluation?
 - ☐ Yes or **No**

Taxation

- Tax should be taken into consideration when calculating the incremental cash flow amounts
- Tax rate is assumed to be 30%
- For example:
 - For a \$1000 cash flow received through sales, we pay 30% of the sales as tax ($\$1,000 \times 30\%$) -> The after-tax cash flow for sales is: $\$1,000 \times (1-30\%) = +\700
 - For a \$500 cash flow paid to expenses, we save 30% of the expense through tax ($\$500 \times 30\%$) -> The after-tax cash flow for expenses is: $-\$500 \times (1-30\%) = -\350
- To simplify the capital budgeting analysis, the incremental cash flow amount should be included to our analysis after taking into account its tax effect.

Taxation

After-tax cash flow = Pre-tax cash flow $\times$ (1 – tax rate)

- We only need to consider tax effects of:
 - o Income, revenue, gain, sales
 - o Expense, cost, loss Income Statement
- No tax effect from:
 - o Asset
 - o Liabilities
 - o Equity Balance Sheet

Depreciation

- **Special case: Purchase of a non-current asset**
 - **No tax effect** on the purchase of a non-current asset. However, a non-current asset is a depreciable asset which means there will be depreciation expense for the asset.
- Does that mean the depreciation expense is included in our evaluation ?
 - **Depreciation expense** is not a cash flow and **is excluded** from the evaluation.
 - **Depreciation expense is tax deductible.** This means it reduces tax expense which is a cash flow.
 - The **depreciation tax saving is included** in the evaluation.

Depreciation

- **Depreciation tax savings = Depreciation $\times$ tax rate**
- In BFIN011 we only use **straight line depreciation**
- **Cost:** The total cost of the non-current asset
- **Asset's useful life:** determined by the Australian Taxation Office (ATO) for tax purposes. This may be different to the project life
 - Asset's useful life: how long we can claim depreciation tax saving for
 - Project life: How long the firm intends to use the machine for
 - The useful life of the asset could be different or similar to the project life. However, an asset's value cannot be depreciated below zero

Practice 2

- Calculate the incremental cash flows that are to be included in our capital budgeting analysis (assuming tax rate is 30%):
 - **Scenario 1:** A decrease in sales of \$10,000
 - A decrease in sales ->a cash outflow of \$10,000
 - After-tax cash flow = $-\$10,000 \times (1 - 30\%) = -\$7,000$
 - **Scenario 2:** An increase in costs of \$20,000
 - A increase in costs ->a cash outflow of \$20,000
 - After-tax cash flow = $-\$20,000 \times (1 - 30\%) = -\$14,000$
 - **Scenario 3:** New inventory worth of \$10,000 is required at the start of the project
 - A increase in current asset -> a cash outflow of \$10,000
 - No tax effect as this is an asset

Practice 2

- **Scenario 4:** A machine with a total cost of \$100,000 is purchased at the start for a 5 year project. According to the Taxation Office, the machine has an useful life of 4 years

The following incremental cash flows are to be included in our analysis:

- The purchase of the machine at the start: -\$100,000
- Depreciation tax saving for the first 4 years: $\frac{100,000}{4} \times 30\% = 7,500$

No depreciation tax saving for the 5th year as the machine is fully depreciated after the 4th year
(Book value =0)

Capital budgeting process

- Incremental after-tax operating cash flows will fall into one of three categories:
 - Start (Year 0)
 - Life (Year 1 $\rightarrow$ Year n)
 - End (Year n)

Cash flow at Start

- Occur at the start of a project's life
 - **Initial cash flows**
 - The preparation year - no operating activities yet
- Includes
 - **Purchase of new equipment** (including installation and delivery if needed)
 - Proceeds from the **sale of old equipment**
 - **Tax on profit/loss** on sale of old equipment
 - **Investment in working capital** (Do we need more/less inventory for the new equipment?)
 - Market value of assets owned that are to be used in the project (**opportunity costs**)

Cash Flows During Life

- Occur during the life of the project
 - **Cash flows from Year 1 to project end (n)**
- Includes:
 - After-tax operating cash flows:
 - Sales
 - Expenses/costs
 - Annual tax saving depreciation (if the asset is not fully depreciated yet)

Cash Flows at End

- Occur during the end of the project
 - Cash flows in final year of project
 - In addition to normal project “life” cash flows at CF_n
- Includes:
 - **Recovery** of investment in working capital
 - **Salvage** (sale) value of assets
 - **Tax on Profit/loss** on sale of assets
- **We assume the firm is restored to the situation before the project was accepted**

Put all together

	Year 0	Year 1	Year n (Last year)
Cash flow items (at start)	\$\$\$	\$\$\$	\$\$\$
Cash flow items (during life)		\$\$\$	\$\$\$
Tax items (Tax expense, tax saving)	\$\$\$	\$\$\$	\$\$\$
Cash flow items (at End)			\$\$\$
Total Net Cash Flow	\$\$\$	\$\$\$	\$\$\$

Some items **(not all)** at start will appear through the life and at the end such as: Opportunity cost (life), Working Capital (end)

Some items during life of the project will still appear in the last year (Sales, operational costs, tax expenses, tax savings). However there are some items that only appear in the last year e.g. sale of machine with its tax effect

Example 1: The Baldwin Company

- ❑ Costs of test marketing (already spent): \$250,000
- ❑ Current market value of proposed factory site (which we own): \$150,000
- ❑ Cost of bowling ball machine: \$100,000 (depreciated according to MACRS 5-year)
- ❑ Increase in net working capital: \$10,000
- ❑ Production (in units) by year during 5-year life of the machine: 5,000, 8,000, 12,000, 10,000, 6,000

The Baldwin Company

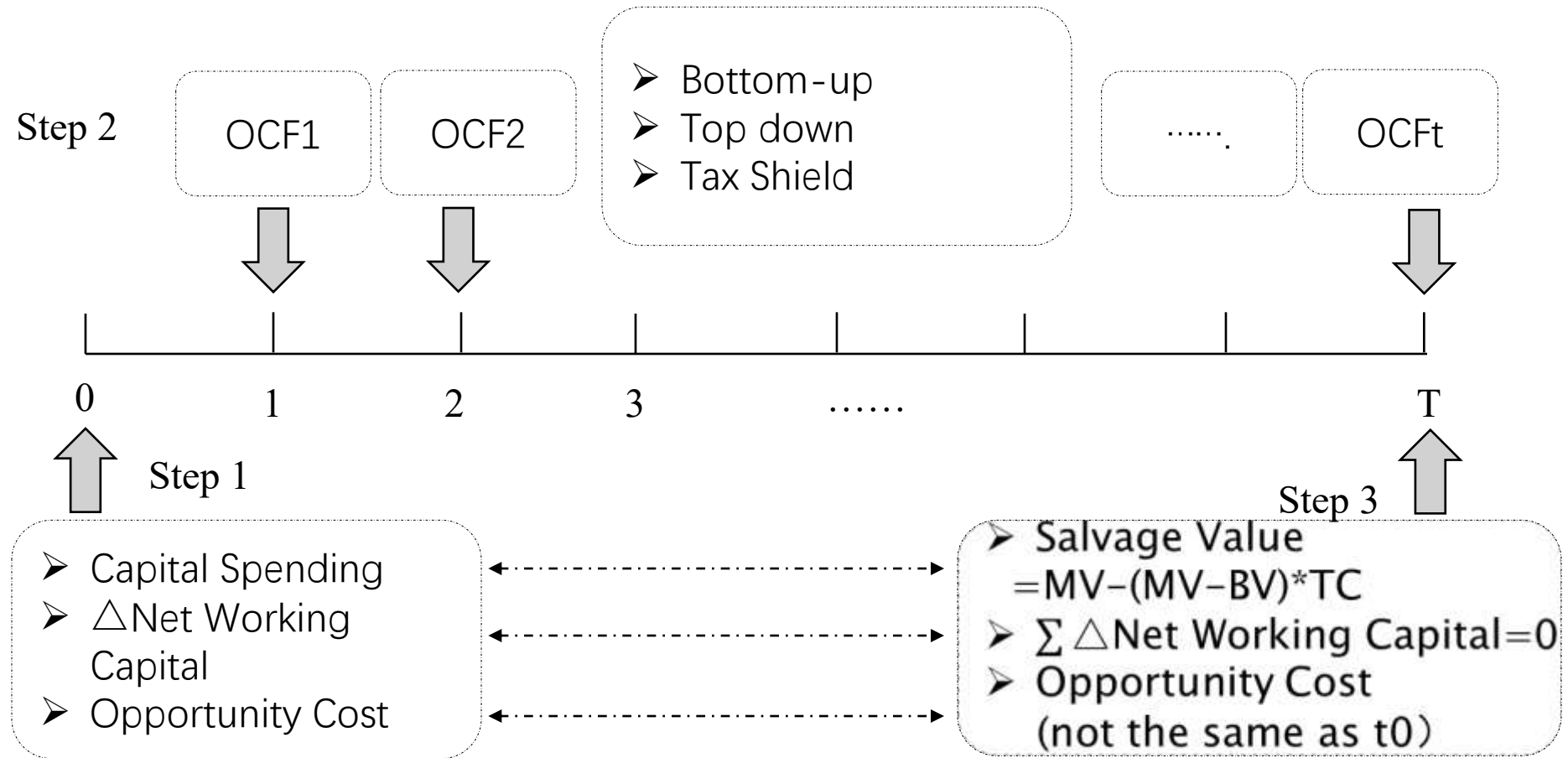
- ❑ Price during first year is \$20; price increases 2% per year thereafter.
- ❑ Production costs during first year are \$10 per unit and increase 10% per year thereafter.
- ❑ Annual inflation rate: 5%
- ❑ Working Capital: initial \$10,000 changes with sales

Depreciation under
Modified Accelerated Cost
Recovery System (MACRS)

RECOVERY PERIOD CLASS						
YEAR	3 YEARS	5 YEARS	7 YEARS	10 YEARS	15 YEARS	20 YEARS
1	.3333	.2000	.1429	.1000	.0500	.03750
2	.4445	.3200	.2449	.1800	.0950	.07219
3	.1481	.1920	.1749	.1440	.0855	.06677
4	.0741	.1152	.1249	.1152	.0770	.06177
5		.1152	.0893	.0922	.0693	.05713
6		.0576	.0892	.0737	.0623	.05285
7			.0893	.0655	.0590	.04888
8			.0446	.0655	.0590	.04522
9				.0656	.0591	.04462
10				.0655	.0590	.04461
11				.0328	.0591	.04462
12					.0590	.04461
13					.0591	.04462
14					.0590	.04461
15					.0591	.04462
16					.0295	.04461
17						.04462
18						.04461
19						.04462
20						.04461
21						.02231

Depreciation is expressed as a percent of the asset's cost. These schedules are based on the IRS Publication 946: *How to Depreciate Property* and other details on depreciation are presented later in the chapter. Note that five-year depreciation actually carries over six years because the IRS assumes purchase is made in midyear.

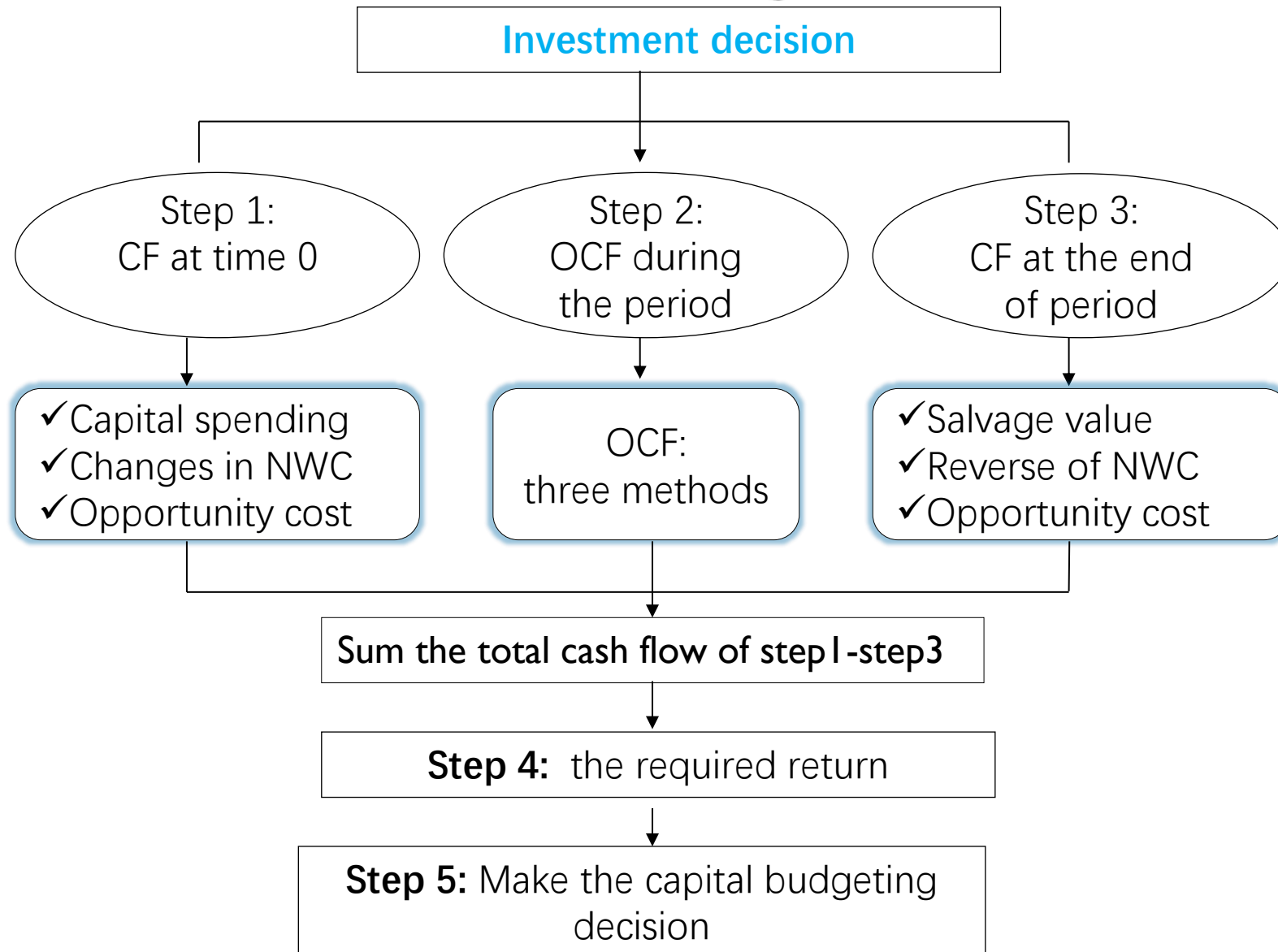
General method of investing decision



Step 4: Calculate the required return

Step 5: Calculate the NPV/IRR/PI/Payback Period/AAR

General method of investing decision



The Baldwin Company

(\$ thousands) (All cash flows occur at the *end* of the year.)

	<i>Year 0</i>	<i>Year 1</i>	<i>Year 2</i>	<i>Year 3</i>	<i>Year 4</i>	<i>Year 5</i>
Investments:						
(1) Bowling ball machine	−100.00					21.76*
(2) Accumulated depreciation		20.00	52.00	71.20	82.72	94.24
(3) Adjusted basis of machine after depreciation (end of year)		80.00	48.00	28.80	17.28	5.76
(4) Opportunity cost (warehouse)	−150.00					150.00
(5) Net working capital (end of year)	10.00	10.00	16.32	24.97	21.22	0
(6) Change in net working capital	−10.00		−6.32	−8.65	3.75	21.22
(7) Total cash flow of investment [(1) + (4) + (6)]	−260.00		−6.32	−8.65	3.75	192.98

At the end of the project, the warehouse is unencumbered, so we can sell it if we want to.

*We assume that the sale price of the bowling ball machine at Year 5 will be \$30 (in thousands). The machine will have been depreciated to \$5.76 at that time. Therefore, the taxable gain from the sale will be \$24.24 ($= \$30 - \5.76). The aftertax salvage value will be $\$30 - [.34 \times (\$30 - \$5.76)] = \21.76 .

The Baldwin Company

	<i>Year 0</i>	<i>Year 1</i>	<i>Year 2</i>	<i>Year 3</i>	<i>Year 4</i>	<i>Year 5</i>
Income:						
(8) Sales Revenues		100.00	163.20	249.72	212.20	129.90

Recall that production (in units) by year during the 5-year life of the machine is given by:

(5,000, 8,000, 12,000, 10,000, 6,000).

Price during the first year is \$20 and increases 2% per year thereafter.

Sales revenue in year 3 = $12,000 \times [\$20 \times (1.02)^2] = 12,000 \times \$20.81 = \$249,720$.

The Baldwin Company

	<i>Year 0</i>	<i>Year 1</i>	<i>Year 2</i>	<i>Year 3</i>	<i>Year 4</i>	<i>Year 5</i>
Income:						
(8) Sales Revenues		100.00	163.20	249.72	212.20	129.90
(9) Operating costs		50.00	88.00	145.20	133.10	87.84

Again, production (in units) by year during 5-year life of the machine is given by:

(5,000, 8,000, 12,000, 10,000, 6,000).

Production costs during the first year (per unit) are \$10, and they increase 10% per year thereafter.

Production costs in year 2 = $8,000 \times [\$10 \times (1.10)^1] = \$88,000$

The Baldwin Company

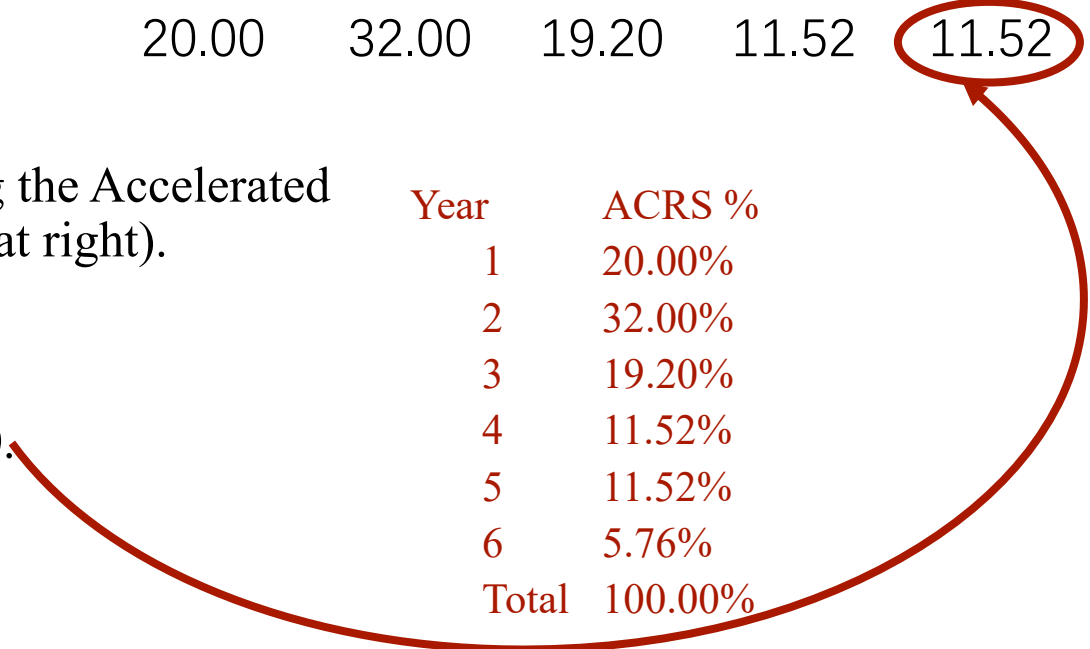
	<i>Year 0</i>	<i>Year 1</i>	<i>Year 2</i>	<i>Year 3</i>	<i>Year 4</i>	<i>Year 5</i>
Income:						
(8) Sales Revenues		100.00	163.20	249.72	212.20	129.90
(9) Operating costs		50.00	88.00	145.20	133.10	87.84
(10) Depreciation		20.00	32.00	19.20	11.52	11.52

Depreciation is calculated using the Accelerated Cost Recovery System (shown at right).

Our cost basis is \$100,000.

Depreciation charge in year 4
= $\$100,000 \times (.1152) = \$11,520$.

Year	ACRS %
1	20.00%
2	32.00%
3	19.20%
4	11.52%
5	11.52%
6	5.76%
Total	100.00%



The Baldwin Company

	<i>Year 0</i>	<i>Year 1</i>	<i>Year 2</i>	<i>Year 3</i>	<i>Year 4</i>	<i>Year 5</i>
Income:						
(8) Sales Revenues	100.00	163.20	249.72	212.20	129.90	
(9) Operating costs	50.00	88.00	145.20	133.10	87.84	
(10) Depreciation	<u>20.00</u>	<u>32.00</u>	<u>19.20</u>	<u>11.52</u>	<u>11.52</u>	
(11) Income before taxes [(8) – (9) - (10)]	<u>30.00</u>	<u>43.20</u>	<u>85.32</u>	<u>67.58</u>	<u>30.54</u>	
(12) Tax at 34 percent	<u>10.20</u>	<u>14.69</u>	<u>29.01</u>	<u>22.98</u>	<u>0.38</u>	
(13) Net Income	<u>19.80</u>	<u>28.51</u>	<u>56.31</u>	<u>44.60</u>	<u>20.16</u>	

Incremental After Tax Cash Flows

	<i>Year 0</i>	<i>Year 1</i>	<i>Year 2</i>	<i>Year 3</i>	<i>Year 4</i>	<i>Year 5</i>
(1) Sales Revenues		\$100.00	\$163.20	\$249.72	\$212.20	\$129.90
(2) Operating costs		-50.00	-88.00	-145.20	133.10	-87.84
(3) Taxes		-10.20	-14.69	-29.01	-22.98	-10.38
(4) OCF		<u>39.80</u>	<u>60.51</u>	<u>75.51</u>	<u>56.12</u>	<u>31.68</u>
(1) – (2) – (3)						
(5) Total CF of Investment	-260.		-6.32	-8.65	3.75	192.98
(6) IATCF [(4) + (5)]	-260.	39.80	54.19	66.86	59.87	224.66

$$NPV = -\$260 + \frac{\$39.80}{(1.10)} + \frac{\$54.19}{(1.10)^2} + \frac{\$66.86}{(1.10)^3} + \frac{\$59.87}{(1.10)^4} + \frac{\$224.66}{(1.10)^5}$$

$$NPV = \$51.588$$

Inflation and Capital Budgeting

- Inflation is an important fact of economic life and might be considered in capital budgeting.
- Consider the relationship between interest rates and inflation, often referred to as the Fisher equation:
$$(1 + \text{Nominal Rate}) = (1 + \text{Real Rate}) \times (1 + \text{Inflation Rate})$$

Inflation and Capital Budgeting

- For low rates of inflation, this is often approximated:

$$\text{Real Rate} \cong \text{Nominal Rate} - \text{Inflation Rate}$$

- ▶ While the nominal rate in the U.S. has fluctuated with inflation, the real rate has generally exhibited far less variance than the nominal rate.
- ▶ In capital budgeting, one must compare real cash flows discounted at real rates or nominal cash flows discounted at nominal rates.

Example 2 :

- Shields Electric forecasts the following nominal cash flows on a particular project:

DATE	0	1	2
CASH FLOW	-\$1,000	\$600	\$650

- The nominal discount rate is 14 percent, and the inflation rate is forecast to be 5 percent. What is the value of the project?

Example 2 :

- **Using Nominal Quantities**

The NPV can be calculated as:

$$\$26.47 = -\$1,000 + \$600/1.14 + \$650/(1.14)^2$$

Alternative Methods for Computing OCF

- Bottom-Up Approach
 - Works only when there is no interest expense
 - $OCF = NI + \text{depreciation}$
- Top-Down Approach
 - $OCF = \text{Sales} - \text{Costs} - \text{Taxes}$
 - Don't subtract non-cash deductions
- Tax Shield Approach
 - $OCF = (\text{Sales} - \text{Costs})(1 - T) + \text{Depreciation} * T$

Example 3:

- For a particular project and year under consideration, suppose we have the following estimates:

Sales \$ 1 , 500

Costs \$ 700

Depreciation \$ 600

Tc 34%

Example 3:

- $\text{EBIT} = \text{Sales} - \text{Costs} - \text{Depreciation}$
 $= \$1,500 - 700 - 600$
 $= \$200$
- $\text{Taxes} = \text{EBIT} \times t_c = \$200 \times .34 = \$68$
- $\text{Project net income} = \text{EBIT} - \text{Taxes}$
 $= \$200 - 68$
 $= \$132$

Example 3:

- **The Bottom-Up Approach**

OCF = Net income + Depreciation

$$=\$ 132 + 600$$

$$=\$ 732$$

- **The Top-Down Approach**

$$\text{OCF} = \text{Sales} - \text{Costs} - \text{Taxes} = \$1,500 - 700 - 68 = \$ 732$$

- **The Tax Shield Approach**

$$\text{OCF} = (\text{Sales} - \text{Costs}) \times (1 - t_c) + \text{Depreciation} \times t_c$$

- $\text{OCF} = (\$ 1,500 - 700) \times .66 + 600 \times .34$

- $= \$ 732$

Example 3:

- **Conclusion**
- all of these approaches are the same
- different approaches are useful in different circumstances.
- The best one to use is whichever happens to be the most convenient for the problem at hand.

6. Project Evaluation Your firm is contemplating the purchase of a new \$625,000 computer-based order entry system. The system will be depreciated straight-line to zero over its five-year life. It will be worth \$60,000 at the end of that time. You will save \$235,000 before taxes per year in order processing costs, and you will be able to reduce working capital by \$55,000 (this is a one-time reduction). If the tax rate is 35 percent, what is the IRR for this project?

6. First, we will calculate the annual depreciation of the new equipment. It will be:

$$\text{Annual depreciation charge} = \$625,000/5$$

$$\text{Annual depreciation charge} = \$125,000$$

The aftertax salvage value of the equipment is:

$$\text{Aftertax salvage value} = \$60,000(1 - .35)$$

$$\text{Aftertax salvage value} = \$39,000$$

Using the tax shield approach, the OCF is:

$$\text{OCF} = \$235,000(1 - .35) + .35(\$125,000)$$

$$\text{OCF} = \$196,500$$

$$\text{NPV} = 0 = -\$625,000 + 55,000 + 196,500(\text{PVIFA}_{\text{IRR}\%,5}) + [(\$39,000 - 55,000)/(1 + \text{IRR})^5] \quad \text{IRR} = 20.90\%$$

- 9. Calculating NPV** Howell Petroleum is considering a new project that complements its existing business. The machine required for the project costs \$3.75 million. The marketing department predicts that sales related to the project will be \$2.1 million per year for the next four years, after which the market will cease to exist. The machine will be depreciated down to zero over its four-year economic life using the straight-line method. Cost of goods sold and operating expenses related to the project are predicted to be 30 percent of sales. The company also needs to add net working capital of \$575,000 immediately. The additional net working capital will be recovered in full at the end of the project's life. The corporate tax rate is 35 percent. The required rate of return is 14 percent. Should the company proceed with the project?

	Equipment	−\$3,750,000
	NWC	<u>−575,000</u>
	Total	−\$4,325,000

	Sales	\$2,100,000
	Costs	630,000
	Depreciation	<u>937,500</u>
	EBT	\$532,500
	Tax	<u>186,375</u>
	Net income	\$346,125

$$\text{OCF} = \text{Net income} + \text{Depreciation}$$

$$\text{OCF} = \$346,125 + 937,500$$

$$\text{OCF} = \$1,283,625$$

$$\text{NPV} = -\$4,325,000 + \$1,283,625(\text{PVIFA}_{14\%,4}) +$$

$$\$575,000/1.14^4$$

$$\text{NPV} = -\$244,439.88$$

22. Calculating NPV and IRR for a Replacement A firm is considering an investment in a new machine with a price of \$9.66 million to replace its existing machine. The current machine has a book value of \$2.52 million, and a market value of \$4.37 million. The new machine is expected to have a four-year life, and the old machine has four years left in which it can be used. If the firm replaces the old machine with the new machine, it expects to save \$2.1 million in operating costs each year over the next four years. Both machines will have no salvage value in four years. If the firm purchases the new machine, it will also need an investment of \$160,000 in net working capital. The required return on the investment is 10 percent and the tax rate is 40 percent.

- a. What are the NPV and IRR of the decision to replace the old machine?
- b. Ignoring the time value of money, the new machine saves only \$8.4 million over the next four years and has a cost of \$9.66 million. How is it possible that the decision to replace the old machine has a positive NPV?

Purchase new machine:

The initial cash outlay for the new machine is the cost of the new machine, plus the increased net working capital. So, the initial cash outlay will be:

	Purchase new machine	−\$9,660,000
	Net working capital	<u>−160,000</u>
	Total	−\$9,820,000

	Operating expense	\$2,100,000
	Depreciation	<u>2,415,000</u>
	EBT	−\$315,000
	Taxes	<u>− 126,000</u>
	Net income	−\$189,000
	OCF	\$2,226,000

$$\text{NPV} = -\$9,820,000 + \$2,226,000(\text{PVIFA}_{10\%,4}) + \$160,000/1.10^4$$

$$\text{NPV} = -\$2,654,597.36$$

$$0 = -\$9,820,000 + \$2,226,000(\text{PVIFA}_{\text{IRR},4}) + \$160,000/(1 + \text{IRR})^4$$

$$\text{IRR} = -3.10\%$$

Keep old machine:

	Keep machine	−\$4,370,000
	Taxes	<u>740,000</u>
	Total	−\$3,630,000

	Depreciation	<u>\$630,000</u>
	EBT	−\$630,000
	Taxes	<u>− 252,000</u>
	Net income	−\$378,000
	OCF	\$252,000

- $NPV = -\$3,630,000 + \$252,000(PVIFA_{10\%,4})$
- $NPV = -\$2,831,193.91$
-
- And the IRR is:
-
- $0 = -\$3,630,000 + \$252,000(PVIFA_{IRR,4})$
-
- $IRR = -36.93\%$

34. Calculating a Bid Price Your company has been approached to bid on a contract to sell 19,000 voice recognition (VR) computer keyboards a year for four years. Due to technological improvements, beyond that time they will be outdated and no sales will be possible. The equipment necessary for the production will cost \$4.9 million and will be depreciated on a straight-line basis to a zero salvage value. Production will require an investment in net working capital of \$325,000 to be returned at the end of the project and the equipment can be sold for \$650,000 at the end of production. Fixed costs are \$1.25 million per year, and variable costs are \$135 per unit. In addition to the contract, you feel your company can sell 8,700, 10,400, 11,700, and 6,700 additional units to companies in other countries over the next four years, respectively, at a price of \$285. This price is fixed. The tax rate is 40 percent, and the required return is 13 percent. Additionally, the president of the company will only undertake the project if it has an NPV of \$500,000. What bid price should you set for the contract?

	<u>Year 1</u>	<u>Year 2</u>	<u>Year 3</u>	<u>Year 4</u>
Sales	\$2,479,500	\$2,964,000	\$3,334,500	\$1,909,500
Variable costs	1,174,500	1,404,000	1,579,500	904,500
EBT	\$1,305,000	\$1,560,000	\$1,755,000	\$1,005,000
Tax	522,000	624,000	702,000	402,000
Net income (and OCF)	\$783,000	\$936,000	\$1,053,000	\$603,000

NPV of market sales = $\$783,000/1.13 + \$936,00/1.13^2 + \$1,053,000/1.13^3 + \$603,000/1.13^4$

NPV of market sales = \$2,525,558.66

Next, we need to calculate the aftertax salvage value, which is:

Aftertax salvage value = $\$650,000(1 - .40)$

Aftertax salvage value = \$390,000

- $NPV = \$500,000 = -\$4,900,000 - 325,000 + 2,525,558.66 + OCF(PVIFA_{13\%,4}) + [(\$325,000 + 390,000)/1.13^4]$
-
- Solving for the OCF, we get:
-
- $OCF = \$2,760,918.45/PVIFA_{13\%,4}$
- $OCF = \$928,204.76$
-
- Now we can solve for the bid price as follows:
-
- $OCF = \$928,204.76 = [(P - v)Q - FC](1 - t_c) + t_c D$
- $\$928,204.76 = [(P - \$135)(19,000) - \$1,250,000](1 - .40) + .40(\$4,900,000/4)$
- $P = \$239.23$
-

36. Project Analysis Benson Enterprises is evaluating alternative uses for a three-story manufacturing and warehousing building that it has purchased for \$2.16 million. The company can continue to rent the building to the present occupants for \$50,000 per year. The present occupants have indicated an interest in staying in the building for at least another 15 years. Alternatively, the company could modify the existing structure and use it for its own manufacturing and warehousing needs. Benson's production engineer feels the building could be adapted to handle one of two new product lines. The cost and revenue data for the two product alternatives are as follows:

	PRODUCT A	PRODUCT B
Initial cash outlay for building modifications	\$114,000	\$132,000
Initial cash outlay for equipment	235,000	282,000
Annual pretax cash revenues (generated for 15 years)	305,000	372,000
Annual pretax expenditures (generated for 15 years)	180,000	230,000

The building will be used for only 15 years for either Product A or Product B. After 15 years, the building will be too small for efficient production of either product line. At that time, Benson plans to rent the building to firms similar to the current occupants. To rent the building again, Benson will need to restore the building to its present layout. The estimated cash cost of restoring the building if Product A has been undertaken is \$65,000. If Product B has been manufactured, the cash cost will be \$78,000. These cash costs can be deducted for tax purposes in the year the expenditures occur.

Benson will depreciate the original building shell over a 30-year life to zero, regardless of which alternative it chooses. The building modifications and equipment purchases for either product are estimated to have a 15-year life. They will be depreciated by the straight-line method. The firm's tax rate is 34 percent, and its required rate of return on such investments is 12 percent.

Assume all cash flows occur at the end of the year. The initial outlays for modifications and equipment will occur today (Year 0), and the restoration outlays will occur at the end of Year 15. Which use of the building would you recommend to management?

We will begin by calculating the NPV of the decision of **continuing to rent the building first.**

Continue to rent:

	Rent	\$50,000
	Taxes	<u>17,000</u>
	Net income	\$33,000

Since there is no incremental depreciation, the operating cash flow is equal to the net income. So, the NPV of the decision to continue to rent is:

$$\text{NPV} = \$33,000(\text{PVIFA}_{12\%,15})$$

$$\text{NPV} = \$224,758.53$$

Product A:

	Building modifications		–\$114,000
	Equipment		<u>–235,000</u>
	Total cash flow		–\$349,000
		<u>Years 1-14</u>	<u>Year 15</u>
	Revenue	\$305,000	\$305,000
	Expenditures	180,000	180,000
	Depreciation	23,267	23,267
	Restoration cost	0	65,000
	EBT	\$101,733	\$36,733
	Tax	34,589	12,489
	NI	\$67,144	\$24,244
	OCF	\$90,411	\$47,511

$$\text{NPV} = -\$349,000 + \$90,411(\text{PVIFA}_{12\%,14}) + \$47,511/1.12^{15}$$

$$\text{NPV} = \$258,937.13$$

- Product B:

	Building modifications	−\$132,000
	Equipment	<u>−282,000</u>
	Total cash flow	−\$414,000

		<u>Years 1-14</u>	<u>Year 15</u>
	Revenue	\$372,000	\$372,000
	Expenditures	230,000	230,000
	Depreciation	27,600	27,600
	Restoration cost	0	78,000
	EBT	\$114,400	\$36,400
	Tax	38,896	12,376
	NI	\$75,504	\$24,024
	OCF	\$103,104	\$51,624

$$\text{NPV} = -\$414,000 + \$103,104(\text{PVIFA}_{12\%,14}) + \$51,624/1.12^{15}$$

$$\text{NPV} = \$278,822.17$$

40. Calculating Project NPV J. Smythe, Inc., manufactures fine furniture. The company is deciding whether to introduce a new mahogany dining room table set. The set will sell for \$6,700, including a set of eight chairs. The company feels that sales will be 2,150, 2,230, 2,560, 2,180, and 1,740 sets per year for the next five years, respectively. Variable costs will amount to 45 percent of sales, and fixed costs are \$3.1 million per year. The new dining room table sets will require inventory amounting to 10 percent of sales, produced and stockpiled in the year prior to sales. It is believed that the addition of the new table will cause a loss of sales of 250 dining room table sets per year of the oak tables the company produces. These tables sell for \$4,900 and have variable costs of 40 percent of sales. The inventory for this oak table is also 10 percent. The company believes that sales of the oak table will be discontinued after three years. J. Smythe currently has excess production capacity. If the company buys the necessary equipment today, it will cost \$11.9 million. However, the excess production capacity means the company can produce the new table without buying the new equipment. The company controller has said that the current excess capacity will end in two years with current production. This means that if the company uses the current excess capacity for the new table, it will be forced to spend the \$11.9 million in two years to accommodate the increased sales of its current products. In five years, the new equipment will have a market value of \$1.68 million if purchased today, and \$7.32 million if purchased in two years. The equipment is depreciated on a seven-year MACRS schedule. The company has a tax rate of 38 percent, and the required return for the project is 14 percent.

- a. Should J. Smythe undertake the new project?
- b. Can you perform an IRR analysis on this project? How many IRRs would you expect to find?
- c. How would you interpret the profitability index?

		<u>Year 1</u>	<u>Year 2</u>	<u>Year 3</u>	<u>Year 4</u>	<u>Year 5</u>
	New	\$14,405,000	\$14,941,000	\$17,152,000	\$14,606,000	\$11,658,000
	Lost sales	–1,225,000	–1,225,000	–1,225,000	0	0
	Total	\$13,180,000	\$13,716,000	\$15,927,000	\$14,606,000	\$11,658,000

The company will have to spend money for inventory in the new table, but will be able to reduce inventory of the existing table. So, the initial cash flow today is:

	New table	–\$1,440,500
	Old table	<u>122,500</u>
	Total	–\$1,318,000

In Year 2, the company will have a cash outflow to pay for the cost of the new equipment. Since the equipment will be purchased in two years rather than now, the equipment will have the higher salvage value. The book value of the equipment in five years will be the initial cost, minus the accumulated depreciation, or:

$$\text{Book value} = \$11,900,000 - 1,700,510 - 2,914,310 - 2,081,310$$

$$\text{Book value} = \$5,203,870$$

The taxes on the salvage value will be:

$$\text{Taxes on salvage} = (\$5,203,870 - 7,320,000)(.38)$$

$$\text{Taxes on salvage} = -\$804,129$$

So, the aftertax salvage value of the equipment in five years will be:

	Sell equipment	\$7,320,000
	Taxes	<u>-804,129</u>
	Total	\$6,515,871

Next, we need to calculate the variable costs each year. The variable costs of the lost sales are included as a variable cost savings, so the variable costs will be:

		<u>Year 1</u>	<u>Year 2</u>	<u>Year 3</u>	<u>Year 4</u>	<u>Year 5</u>
	New	\$6,482,250	\$6,723,450	\$7,718,400	\$6,572,700	\$5,246,100
	Lost sales	−490,000	−490,000	−490,000	0	0
	Total	\$5,992,250	\$6,233,450	\$7,228,400	\$6,572,700	\$5,246,100

		<u>Year 1</u>	<u>Year 2</u>	<u>Year 3</u>	<u>Year 4</u>	<u>Year 5</u>
	Sales	\$13,180,000	\$13,716,000	\$15,927,000	\$14,606,000	\$11,658,000
	VC	5,992,250	6,233,450	7,228,400	6,572,700	5,246,100
	Fixed costs	3,100,000	3,100,000	3,100,000	3,100,000	3,100,000
	Dep.	0	0	1,700,510	2,914,310	2,081,310
	EBT	\$4,087,750	\$4,382,550	\$3,898,090	\$2,018,990	\$1,230,590
	Tax	1,553,345	1,665,369	1,481,274	767,216	467,624
	NI	\$2,534,405	\$2,717,181	\$2,416,816	\$1,251,774	\$762,966
	Dep.	0	0	1,700,510	2,914,310	2,081,310
	OCF	\$2,534,405	\$2,717,181	\$4,117,326	\$4,166,084	\$2,844,276

3. This is an opportunity cost. The inventory each year, and the inventory change, will be:

	Beginning	\$1,440,500	\$1,494,100	\$1,715,200	\$1,460,600	\$1,165,800
	Ending	1,494,100	1,715,200	1,460,600	1,165,800	0
	Existing	0	0	−122,500	0	0
	Change	−\$53,600	− \$221,100	\$132,100	\$294,800	\$1,165,800 \$ 1,165,800
		<u>Year 1</u>	<u>Year 2</u>	<u>Year 3</u>	<u>Year 4</u>	<u>Year 5</u>
	OCF	\$2,534,405	\$2,717,181	\$4,117,326	\$4,166,084	\$2,844,276
	Equipment	0	−11,900,000	0	0	6,515,871
	Inventory	−\$53,600	− \$221,100	\$132,100	\$294,800	1,165,800 \$ 1,165,800
	Total	\$2,480,805	−\$9,403,919	\$4,249,426	\$4,460,884	\$10,525,946

$$\begin{aligned}\text{NPV} &= -\$1,318,000 + \$2,480,805/1.14 - \$9,403,919/1.14^2 + \\ &\$4,249,426/1.14^3 + \$4,460,884/1.14^4 + \$10,525,946/1.14^5 \\ \text{NPV} &= \$4,598,423.84\end{aligned}$$

- b.* You can perform an IRR analysis, and would expect to find up to three IRRs since the cash flows change signs three times.
- c.* The profitability index is intended as a “bang for the buck” measure; that is, it shows how much shareholder wealth is created for every dollar of initial investment. In this case, the largest investment is not at the beginning of the project, but later in its life. However, since the future negative cash flow is discounted, the profitability index will still measure the amount of shareholder wealth created for every dollar spent today.